Studio MM S.r.l.
Consulenza materie prime - Prove materiali
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Client:	***IRON TRADE S.r.l.***

Plastering/rendering analysis in accordance with standard UNI EN 998-1

Materials:	*"Thermolock" mortar*

Acceptance:	*RLM_136-2023*

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DETERMINATION OF BULK DENSITY OF HARDENED AND DRIED MORTAR (UNI EN 1015-10:2007)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Test conducted on	17 March 2023
Test report no.	**RLM_126-2023/MV1**

Moulding (date and time)	17 February 2023 at 10:30 AM
Mortar consistence (mm)	148
Type of specimens	Prismatic samples 16 x 4 x 4 mm
Curing	6 days at 20°C and R.H. of 95 ± 5% + 21 days at 20°C and R.H. of 65 ± 5%
Conditioning of test specimens	Samples dried in oven at 105 ± 5°C up to constant mass

Specimen identifier	**Specimen dry mass $M_{s,dry}$ (kg)**	**Nominal volume of the specimen V_s (m^3)**	**Bulk density (kg/m^3)**
Specimen 1	0.1513	256×10^{-6}	591
Specimen 2	0.1488	256×10^{-6}	581
Specimen 3	0.1502	256×10^{-6}	587
		Mean value	**586**

The Technician

Umberto Lorenzi

The Technical Department

Instruments and equipment used for the test	Radwag balance (int. code BL07-C)

Note

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FLEXURAL STRENGTH OF PLASTERING/RENDERING MORTAR SPECIMENS (UNI EN 1015-11:2019)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar; prismatic specimens made in 16 x 4 x 4 cm moulds via impact compactor (2 layers, 25 blows per layer), water/mortar ratio = 0.75 (figure given by the Client)
Test conducted on	15 March 2022
Test report no.	**RLM_126-2023/FL1**

Moulding (date and time)	15 February 2023 at 10:30 AM
Mortar consistence (mm)	146
Curing	7 days at 20°C and R.H. of 95 ± 5% + 21 days at 20°C and R.H. of 65 ± 5%
Loading rate applied	30 N/s

Specimen identifier	**Maximum load applied (N)**	**Distance between supporting rollers (mm)**	**Lat. Size of specimen d_1 d_2 ^{2} (mm)**	**Flexural strength (Mpa)**
Specimen 1	329	100	40	**0.77**
Specimen 2	335	100	40	**0.79**
Specimen 3	317	100	40	**0.74**
			Mean result	**0.77**

The Technician	**The Technical Department**
Umberto Lorenzi	

Instruments and equipment used for the test	MATEST multispeed load frame, int. code PMS01

Note

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COMPRESSION TESTING OF PLASTERING/RENDERING MORTAR SPECIMENS (UNI EN 1015-11:2019)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar; prismatic specimens made in 16 x 4 x 4 cm moulds via impact compactor (2 layers, 25 blows per layer), water/mortar ratio = 0.75 (figure given by the Client)
Test conducted on	15 March 2023
Test report no.	**RLM_136-2023/RC1**

Date and time of moulding	15 February 2023 at 10:30 AM
Mortar consistence (mm)	146
Curing	7 days at 20°C and R.H. of 95 ± 5% + 21 days at 20°C and R.H. of 65 ± 5%
Loading rate applied	50 N/s

Specimen identifier		**1a**	**1b**	**2a**	**2b**	**3a**	**3b**
Specimen type		Piece of broken prismatic specimen	Piece of broken prismatic specimen	Piece of broken prismatic specimen	Piece of broken prismatic specimen	Piece of broken prismatic specimen	Piece of broken prismatic specimen
Surface condition		regular	regular	regular	regular	regular	regular
Curing		*28 days*	*28 days*	*7 days*	*28 days*	*28 days*	*28 days*
Sizes (mm)	a	40	40	40	40	40	40
	b	40	40	40	40	40	40
	h	40	40	40	40	40	40
Area (mm^2)		1600	1600	1600	1600	1600	1600
Maximum load at breaking (N)		2924	2918	2880	2975	2885	2900
Compressive strength (MPa)		**1.83**	**1.82**	**1.80**	**1.86**	**1.80**	**1.81**

Compressive strength, mean value (MPa)	**1.82**

The Technician Umberto Lorenzi	**The Technical Department**

Notes:

Instruments and equipment used for the test	MATEST multispeed load frame, int. code PMS01

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DETERMINATION OF ADHESIVE STRENGTH OF PLASTERING/RENDERING MORTAR ON SUBSTRATES (UNI EN 1015-12:2016)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Test date and time	28 March 2023 at 11:00 AM
Test report no.	**RLM_126-2023/PO1**

Moulding (date and time)	1 March 2023, at 10:30 AM
Mortar consistence (mm)	146
Substrate type	Concrete panel, 600 x 150 x 150 mm, water-cement ratio 0.55
Panel orientation	horizontal
Plastering/rendering layer thickness	10 mm
Sample curing	in PVC bag for 7 days at 20°C; exposed to air for 21 days at 20°C and R.H. 65±5 %

Sample identifier	**Breaking load *Fu* (N)**	**Test area *A* (mm^2)**	**Adhesion strength *fu* (N/mm^2)**	**Type of break**
Sample 1	530	1962.5	0.27	Cohesion fracture in the mortar itself
Sample 2	399	1962.5	0.20	Cohesion fracture inside the mortar
Specimen 3	466	1962.5	0.24	Cohesion fracture in the mortar itself
Sample 4	502	1962.5	0.26	Cohesion fracture in the mortar itself
Specimen 5	438	1962.5	0.22	Cohesion fracture in the mortar itself
		Mean value	**0.24**	

The Technician	**The Technical Department**
Umberto Lorenzi	

Instruments and equipment used for the test	Matest Pull Off Tester (int. code POF01-T)

Note

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DETERMINATION OF WATER ABSORPTION COEFFICIENT DUE TO CAPILLARY ACTION OF HARDENED MORTAR (UNI EN 1015-18:2002)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Test conducted on	28 March 2023
Test report no.	**RLM_126-2023/AC1**

Moulding (date and time)	1 March 2023, at 11:30 AM
Mortar consistence (mm)	145
Type of specimens	Prismatic samples 16 x 4 x 4 mm each one broken in two pieces
Curing	7 days at 20°C and R.H. of 95 ± 5% + 21 days at 20°C and R.H. of 65 ± 5%
Conditioning of test specimens	Samples dried in oven at 60 ± 5°C up to constant mass

Specimen identifier	**Specimen mass after 10 min of immersion M1 (g)**	**Specimen mass after 10 min of immersion M2 (g)**	**Absorption coefficient C (kg/(m^2 x min ^{0,5}))**
Specimen 1a	85.1	89.6	0.45
Specimen 1b	79.8	83.9	0.41
Specimen 2a	84.0	87.8	0.38
Specimen 2b	78.1	82.1	0.40
Specimen 3a	84.5	88.6	0.41
Specimen 3b	81.6	86.0	0.44
		Mean value C_m	**0.42**

The Technician	**The Technical Department**
Umberto Lorenzi	

Instruments and equipment used for the test	Radwag balance (int. code BL07-C)

Note

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Determination of water vapour permeability of hardened rendering and plastering mortars (UNI EN 1015-19:2008)

Test report no.	RLM_136-2023_PV01
Client	Iron Trade S.r.l.
Sample description	"Thermolock" plastering/rendering mortar
Sample identification	"Thermolock" plastering/rendering mortar
Test conducted	From 3 May 2021 to 19 May 2023

TEST RESULT

Test temperature (C°)	20
Specimen average thickness (m)	0.02
Opening area (m^2)	0.02
Partial vapour pressure difference KNO3 (Pa)	1011
Partial vapour pressure difference LICl (Pa)	879
Relative humidity at 20° KNO3	93.20%
Relative humidity at 20° LICl	50%

Test conducted in **high relative humidity** conditions, vapour tension inside the sample-holder maintained using a saturated solution in **KNO3**

Test sample	Specimen thickness (m)	Specimen mass (g)	Vapour flow (Kg/s)
Specimen 1	0.02275	485.8	-1,63E-08
Specimen 2	0.02325	495.2	-1.63E-08
Specimen 3	0.02325	482.3	-1.59E-08
Specimen 4	0.023	484.1	-1.67E-08
Specimen 5	0.023	486.3	-1.55E-08

Tests conducted: from 3 May to 17 May

Test conducted in **low relative humidity** conditions, vapour tension inside the sample-holder maintained using a saturated solution in **LICl**

Test sample	Specimen thickness (m)	Specimen mass (g)	Vapour flow (Kg/s)
Specimen 1	0.02	485.4	1.02E-08
Specimen 2	0.0194	480.4	1.02E-08
Specimen 3	0.02	482.4	1.02E-08
Specimen 4	0.0198	481.4	1.02E-08
Specimen 5	0.0198	483.2	1.02E-08

Tests conducted: from 22 June to 8 July

Average permeability	8,3059E-10
Average permeability to vapour (Wvp)	1,91E-11
Average coefficient of resistance to vapour diffusion (m) high relative humidity	10.1
Average coefficient of resistance to vapour diffusion (m) low relative humidity	16.3
Average coefficient of resistance to vapour diffusion (m)	**13.2**

The Technician	**The Technical Department**
Dott.ssa Angela Dettori	Studio MM s.r.l. Consulenza materie prime e prove materiali Dott. Geol. Mazzoni Michele

Instruments and equipment used for the test: The testing equipment is compliance with standard UNI EN 1015-19:2008

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CHEMICAL ANALYSIS (UNI EN 196-2)

Client	Iron Trade S.r.l.
Material	"Thermolock" plastering/rendering mortar
Test report no.	RLM_136-2023/SC1
Test conducted	28 February 2023

GRAVIMETRIC DETERMINATION OF SULPHATES EXPRESSED AS SO_3 (UNI EN 196-2 Para. 8)	
Sulphate content (SO_3) =	**0.86%**

The Technician	**The Technical Department**
Eleonora Colla	Studio tecnologico M & M Consulenza materie prime e prove materiali Dott. Geol. Mazzoni Michele

Notes:

Instruments and equipment used for the test:	Radwag balance (internal code BL-04-T)

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DETERMINATION OF THE COMPATIBILITY OF ONE-COAT RENDERING MORTARS WITH SUBSTRATES - WATER PERMEABILITY AND ADHESION STRENGTH (UNI EN 1015-21:2004)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Test conducted on	18 April - 11 May 2023
Test report no.	**RLM_126-2023/GD1**

Moulding (date and time)	6 March 2023, at 9:30 AM
Mortar consistence (mm)	145
Substrate type	Concrete panels 300x300x50 mm and brick panels 400 x 400 mm
Curing	28 days at 20°C and R.H. of 65 ± 5%

Sample conditioning:
4 heating cycles (infrared radiation 60°C ± 2°C) - freezing (-15°C ± 1°C)
4 humidification cycles (20°C ± 2°C) - freezing (-15°C ± 1°C)

WATER PERMEABILITY AFTER CYCLES OF EXPOSURE TO WEATHER

Substrate type	Plastering/rendering thickness (mm)	Permeability ml/cm^2 after 48h
Concrete panel	10	0.95
Concrete panel	20	0.96
Brick panel	10	0.99
Brick panel	20	0.99

Notes

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ADHESION STRENGTH TO SUBSTRATE AFTER EXPOSURE TO WEATHERING CYCLES (UNI EN 1015-21:2004)

Specimen identifier	Breaking load *Fu* (N)	Test area *A* (mm^2)	Bond strength *fu* (N/mm^2)	Type of break
Concrete substrate; thickness 1mm	301	1962.5	0.15	Cohesion fracture in the mortar itself
Concrete substrate; thickness 1 mm	355	1962.5	0.18	Cohesion fracture in the mortar itself
Concrete substrate; thickness 1 mm	402	1962.5	0.20	Cohesion fracture in the mortar itself
Concrete substrate; thickness 1 mm	312	1962.5	0.16	Cohesion fracture in the mortar itself
Concrete substrate; thickness 1 mm	307	1962.5	0.16	Cohesion fracture in the mortar itself
		Mean value	**0.17**	

Specimen identifier	Breaking load *Fu* (N)	Test area *A* (mm^2)	Adhesion strength *fu* (N/mm^2)	Type of break
Concrete substrate; thickness 2 mm	298	1962.5	0.15	Cohesion fracture in the mortar itself
Concrete substrate; thickness 2 mm	388	1962.5	0.20	Cohesion fracture in the mortar itself
Concrete substrate; thickness 2 mm	379	1962.5	0.19	Cohesion fracture in the mortar itself
Concrete substrate; thickness 2 mm	405	1962.5	0.21	Cohesion fracture in the mortar itself
Concrete substrate; thickness 2 mm	377	1962.5	0.19	Cohesion fracture in the mortar itself
		Mean value	**0.19**	

Sample identifier	Breaking load *Fu* (N)	Test area *A* (mm^2)	Adhesion strength *fu* (N/mm^2)	Type of break
Brick substrate; thickness 1mm	388	1962.5	0.20	Cohesion fracture in the mortar itself
Brick substrate; thickness 1mm	376	1962.5	0.19	Cohesion fracture in the mortar itself
Brick substrate; thickness 1mm	251	1962.5	0.13	Cohesion fracture in the mortar itself
Brick substrate; thickness 1 mm	284	1962.5	0.14	Cohesion fracture in the mortar itself
Brick substrate; thickness 1 mm	293	1962.5	0.15	Cohesion fracture in the mortar itself
		Mean value	**0.16**	

Sample identifier	Breaking load *Fu* (N)	Test area *A* (mm^2)	Adhesion strength *fu* (N/mm^2)	Type of break
Brick substrate; thickness 2 mm	390	1962.5	0.20	Cohesion fracture in the mortar itself
Brick substrate; thickness 2 mm	405	1962.5	0.21	Cohesion fracture in the mortar itself
Brick substrate; thickness	378	1962.5	0.19	Cohesion fracture in the mortar itself

2 mm				
Brick substrate; thickness 2 mm	372	1962.5	0.19	Cohesion fracture in the mortar itself
Brick substrate; thickness 2 mm	366	1962.5	0.19	Cohesion fracture in the mortar itself
		Mean value	**0.19**	

The Technician	**The Technical Department**
Umberto Lorenzi	

MASONRY AND MASONRY PRODUCTS - DETERMINATION OF THE PROJECT THERMAL VALUES (EN 1745:2012)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Report date	2 May 2023
Report no.	**RLM_136-2023/CT1**

Material bulk density	**λ10 dry**		**Water vapour resistance factor μ**	**Specific thermal capacity C_p**
kg/m³	**W/(mK)**			**J/(kgK)**
	***P* 50%**	***P* 90%**		
600	**0.14**	**0.15**	**5/20**	**1000**

The Technician

Umberto Lorenzi

The Technical Department

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DETERMINATION OF THERMAL PERFORMANCE
GUARDED HOT PLATE METHOD (UNI EN 12664-2002)

Client	Iron Trade S.r.l.
Sampling	By the Client
Sample description	"Thermolock" plastering/rendering mortar
Test conducted on	17-23 May 2023
Test report no.	**RLM_136-2023/CT2**

Foreword: thermal performance was determined on 3 specimens with sizes 250x250x50 mm using the guarded hot plate method on a single specimen; the instrument has vertical orientation and the hot side of the sample is the upper one. The test was conducted in a conditioned chamber at 23±2°C and relative humidity of 50±5%. The variation in the thermal conductivity parameter in the stable part of the test is lower than 1% over duration of 480 minutes.

Conditioning: the specimens were conditioned in a ventilated oven until constant mass was reached. Bulk density before conditioning was 650 kg/m^3, bulk density at the end of conditioning was 619 kg/m^3

Sample	**Temperature difference through the specimen**	**Heat flow density through the specimen**	**Thermal conductivity**	**Thermal resistance**
	ΔT	q	λ_{10}	R_{10}
	K	W/m^2	W/(m*K)	(m^2*K)/W
"Thermolock" plastering/rendering mortar	15	42.89	**0.147**	0.35

Instruments and equipment used:	Guarded hot plate Lambda Meter EP 500, Lambda Messtechnik GmbH Dresden

The Technician	**The Technical Department**
Umberto Lorenzi	